Ex.No. 12:

NB-IoT vs LTE-M Throughput Simulation Date

AIM:

To simulate and compare the performance of NB-IoT and LTE-M communication technologies in terms of Throughput (Mbps), Latency (ms), and Packet Delivery Ratio (%) using MATLAB.

Objective:

1. To evaluate and compare the Throughput (Mbps) performance of NB-IoT and LTE-M under varying network conditions.

2. To analyze the Latency (ms) characteristics of NB-IoT and LTE-M and study their impact on real-time communication.

3. To measure and compare the Packet Delivery Ratio (%) of NB-IoT and LTE-M systems for reliability assessment.

Objective 1 Matlab Code:

```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Throughput (Mbps)

NB_IoT = 0.05 * (1 - exp(-SNR/8));

LTE_M  = 1.2  * (1 - exp(-SNR/8));

disp('SNR (dB)    NB-IoT(Mbps)    LTE-M(Mbps)')

disp([SNR' NB_IoT' LTE_M'])

figure;

% -------- Graph 1 --------

subplot(2,1,1)

plot(SNR, NB_IoT, '-o', 'LineWidth', 2)

title('NB-IoT Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on

% -------- Graph 2 --------

subplot(2,1,2)
```

plot(SNR, LTE_M, '-s', 'LineWidth', 2)

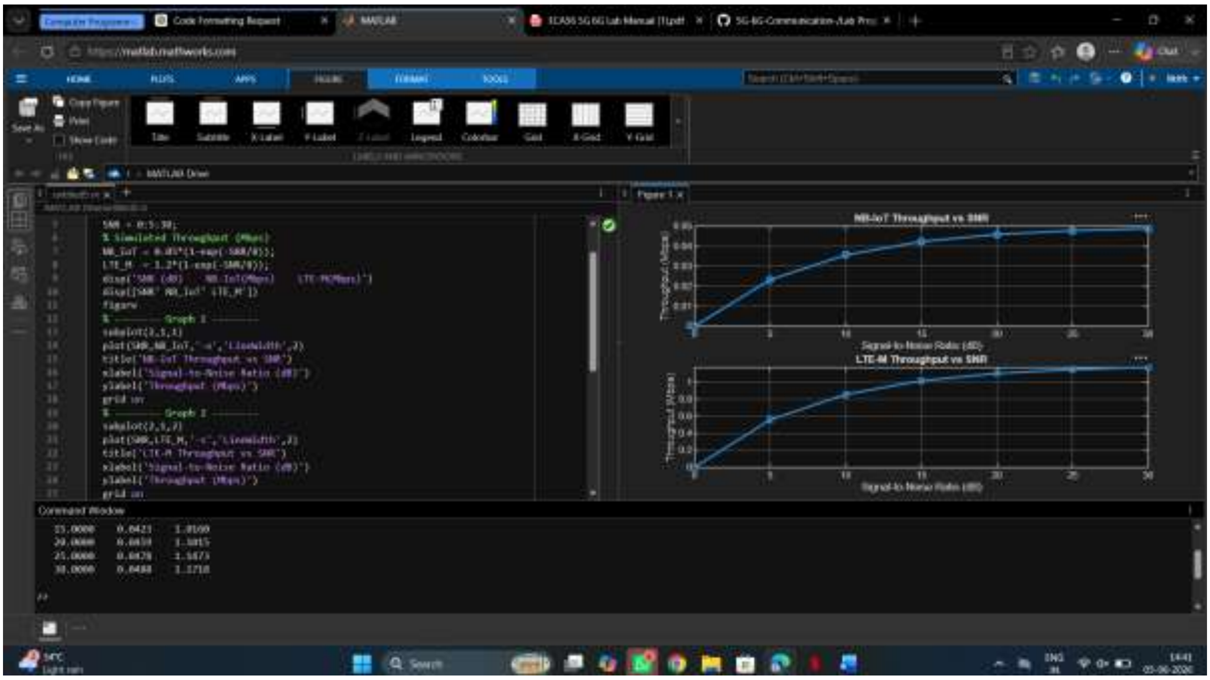
title('LTE-M Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on

OUTPUT :



Objective 2 Matlab Code :

clc;

clear;

close all

% SNR values (dB)

SNR = 0:5:30;

% Simulated Latency (ms)

NB_IoT = [180 160 145 130 120 110 100];

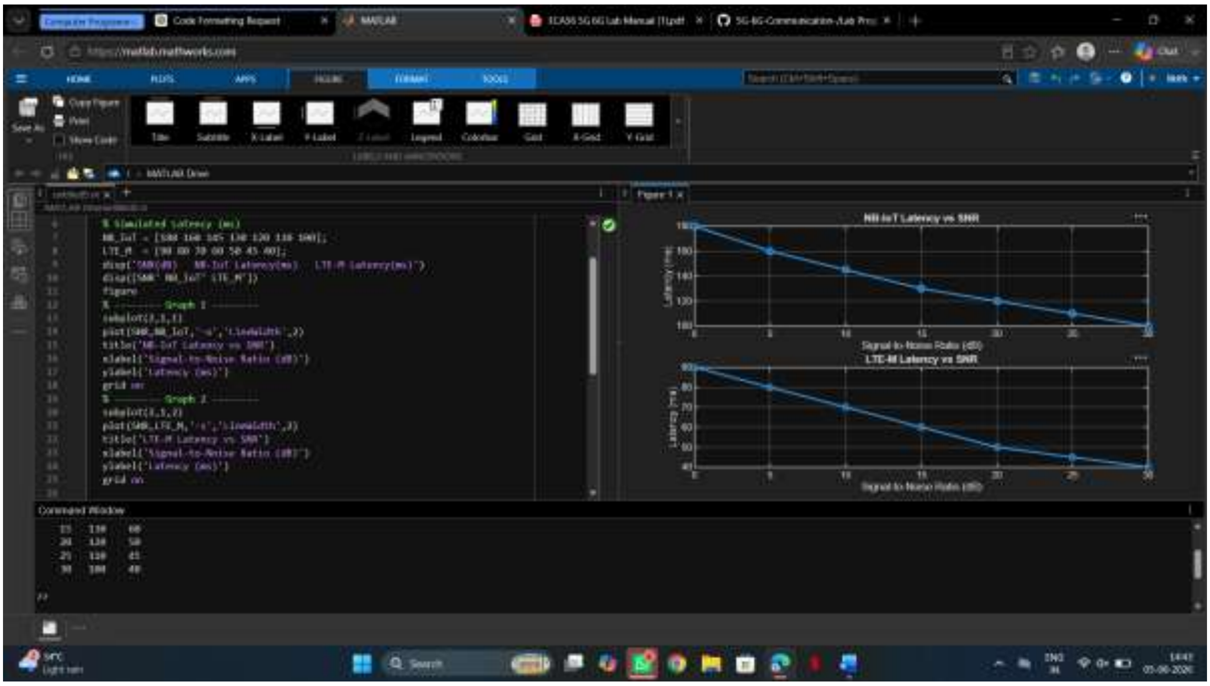
LTE_M = [90 80 70 60 50 45 40]

disp('SNR(dB)   NB-IoT Latency(ms)   LTE-M Latency(ms)')

disp([SNR' NB_IoT' LTE_M'])

```
figure;
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR, NB_IoT, '-o', 'LineWidth', 2)
title('NB-IoT Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR, LTE_M, '-s', 'LineWidth', 2)
title('LTE-M Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
```

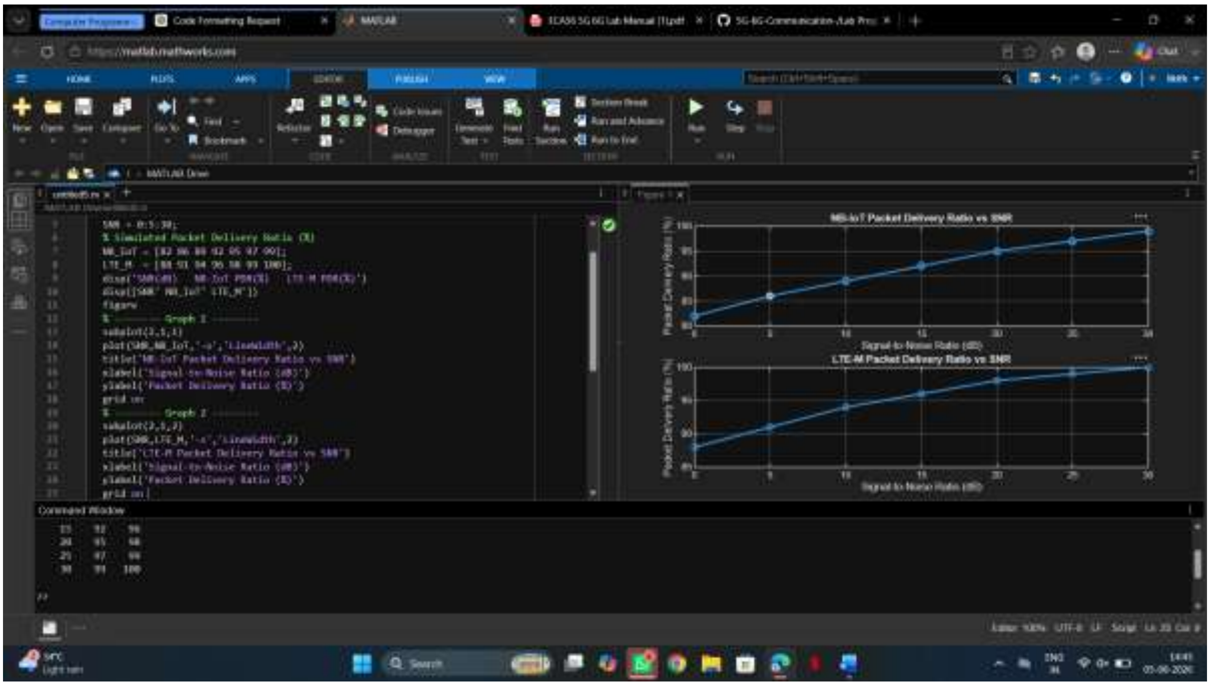
OUTPUT :



Objective 3 Matlab Code ;

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Packet Delivery Ratio (%)
NB_IoT = [82 86 89 92 95 97 99];
LTE_M  = [88 91 94 96 98 99 100];
disp('SNR(dB)   NB-IoT PDR(%)   LTE-M PDR(%)')
disp([SNR' NB_IoT' LTE_M'])
figure;
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR, NB_IoT, '-o', 'LineWidth', 2)
title('NB-IoT Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR, LTE_M, '-s', 'LineWidth', 2)
title('LTE-M Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
```

OUTPUT :



Result:

1. The throughput performance of both NB-IoT and LTE-M was successfully simulated, and LTE-M achieved higher throughput than NB-IoT under the same network conditions.

2. The latency analysis showed that LTE-M provides lower communication delay compared to NB-IoT, making it more suitable for real-time IoT applications.

3. The Packet Delivery Ratio (PDR) of both technologies improved with increasing SNR, with LTE-M demonstrating slightly higher reliability than NB-IoT during data transmission.